SPARK: Spectral Post-Attention Repair Kernels for Underwater Image Enhancement

BMVC 2026 Submission # 301

Abstract

Underwater images degrade at two coupled frequency scales: low-frequency global appearance is distorted by colour cast, haze, illumination drift, and visibility loss, while high-frequency local structures such as edges, textures, and fine details are attenuated. Transformer-style UIE backbones improve global context aggregation, but their feed-forward networks (FFNs) often remain generic mixers rather than restoration-specific operators. Frequency-aware processing is therefore a natural design direction, yet its insertion point inside a restoration block remains underexplored. We argue that frequency separation should happen after attention: attention first contextualizes features with scene-level evidence, and the following FFN can then translate this global view into band-specific repair. Based on this principle, we propose SPARK (Spectral Post-Attention Repair Kernels), a post-attention FFN that replaces only the original post-attention unit and follows a compact split–repair–fuse–refine design over low- and high-frequency components. Under the matched 100-epoch ablation, this post-attention instantiation improves a strong gated baseline and gives the best three-domain mean on PSNR, SSIM, LPIPS, and UCIQE among the controlled placement variants, while pre-attention substitutions and routing/fusion variants fail to match it. These results indicate that frequency awareness alone is not enough for underwater image enhancement; its placement inside the restoration block is a first-order design decision.

1 Introduction

Underwater images rarely degrade in a single way. As light travels through water, wavelength-dependent attenuation and scattering change the image slowly at the scene level, producing colour casts, veiling haze, illumination drift, and reduced visibility [1, 4, 12, 16]. At the same time, turbidity and backscattering suppress local evidence, weakening edges, washing out textures, and attenuating fine structures that are important for perception. These two effects are coupled rather than independent: the same image may require global appearance correction while also needing local detail recovery. Underwater image enhancement (UIE) is therefore not simply a colour correction problem, nor merely a sharpening problem. It is a frequency-asymmetric restoration task, where low-frequency appearance degradations and high-frequency structure degradations must be addressed together.

This two-scale view also clarifies what recent UIE progress has solved, and what it has not. Earlier prior-based and physics-guided methods have been largely complemented by data-driven models, with Water-Net and UIEB making supervised real-image evaluation more systematic, Ucolor injecting medium-transmission and multi-colour-space cues,

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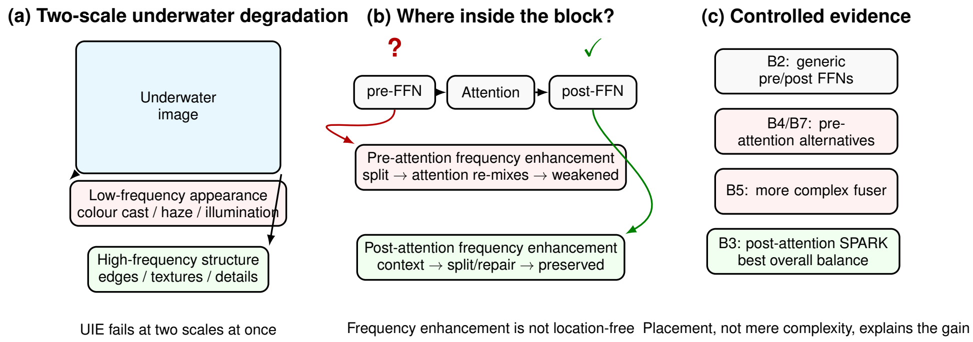

Figure 1: **Where should frequency enhancement happen?** Underwater image enhancement must jointly address slow-varying appearance degradations and local structural degradations. Frequency-aware processing is therefore natural, but not location-free: pre-attention operations can be diluted when attention re-mixes spatial information, whereas post-attention FFNs operate on context-aware features and can convert the global view into band-specific repair.

and U-Shape Transformer introducing transformer-style global modeling for spatially and channel-wise non-uniform underwater attenuation [16, 17, 19]. These developments show that UIE benefits from stronger context aggregation. However, transformer-style global modeling helps, but global colour correction and local detail preservation are not automatically aligned. A model may reduce colour cast and haze while over-smoothing textures, or preserve sharp structures while leaving large-scale visibility and illumination errors unresolved. Robust UIE therefore depends not only on building a stronger global backbone, but on how the restoration block couples scene-level appearance correction with local structural recovery.

This mismatch brings us from the backbone level to the block level. In transformer-style restoration networks, attention is usually credited with aggregating long-range context, but the feed-forward network (FFN) is where the representation is locally transformed into a restoration update. Recent low-level vision models have therefore begun to revisit the FFN itself, replacing the naive MLP-style mixer with gated, convolutional, or frequency-aware variants that better match image restoration [11, 14, 23]. This view is particularly useful in a block that places one FFN before attention and another after it. The two slots expose a simple but underused design axis. If attention provides the global view, the surrounding FFNs determine how that view is converted into appearance correction and detail recovery. In this sense, the FFN is not a fixed default component, but an overlooked restoration operator.

This design axis is where frequency-aware restoration becomes more than a choice of representation. For UIE, separating low- and high-frequency information is natural: slow-varying colour shift, haze, and illumination drift should not be processed in the same way as edges, textures, and fine structures. However, existing UIE architectures usually introduce such cues through full-network redesigns, auxiliary branches, or attention-level modifications, making it difficult to isolate placement inside a single restoration block. As illustrated in Figure 1, the central question is therefore not whether frequency-aware enhancement is useful, but where it should be inserted: before attention, when features have not yet been globally mixed, or after attention, when the FFN can act on context-aware representations.

We hypothesize that frequency separation should happen after attention. The pre-attention FFN operates before long-range context has been aggregated, and its output is immediately passed into self-attention, where spatial information is re-mixed across the feature map. Frequency-specific operations inserted at this stage may therefore be weakened before they can act as a stable restoration update. By contrast, the post-attention FFN receives features that have already collected global evidence about colour bias, haze distribution, illumination, and object structure. This makes it a more suitable location for converting the global view into band-specific repair: low-frequency components can be used to correct slow-varying appearance degradations, while high-frequency residuals can focus on edges, textures, and fine structures. We therefore keep the pre-attention FFN as a standard gated feed-forward unit and redesign only the post-attention FFN into SPARK.

To test this hypothesis, we introduce SPARK as a slot-specific redesign of the post-attention FFN, rather than as a full backbone replacement. The module follows a simple restoration logic: split, repair, fuse, and refine. It first separates the context-aware feature into low- and high-frequency components. The low-frequency branch targets slow-varying appearance degradations such as colour bias, haze, and illumination drift, while the high-frequency branch focuses on local evidence such as edges, textures, and fine structures. The two streams are then adaptively fused so that the model can choose different low/high balances for different images and channels, and a gated feed-forward refinement converts the fused representation into a stable restoration update. In this way, SPARK is not intended to make the whole network more complicated; it is a targeted intervention at the position where frequency-aware repair is most likely to survive.

Our experiments are organized to test this placement hypothesis rather than to present SPARK as a monolithic new backbone. Under matched 100-epoch training, the post-attention design improves a strong gated/gated baseline and gives the best overall balance across PSNR, SSIM, LPIPS, and UCIQE among the controlled placement variants, with only a small UIQM gap to a more complex pre-attention variant. More importantly, several plausible alternatives fail to explain the gain: modifying the pre-attention FFN with deeper gated mixing does not match B3, a pre-attention frequency MoE remains below B3, and a more complex channel-spatial fuser inside the post-attention design also falls short. These comparisons make the result diagnostic rather than merely incremental.

Contributions. Our main contributions are as follows.

- **Problem.** We identify FFN slot placement as an important but underexplored design factor for frequency-aware underwater image enhancement. Rather than asking only whether frequency decomposition helps, we isolate where it should be inserted inside a Transformer-style restoration block and study the functional difference between pre- and post-attention FFN slots.

- **Method and controlled implementation.** We implement our own compact B-series UIE architecture as a controlled testbed for FFN-slot placement, and propose SPARK, a slot-specific post-attention FFN that performs content-adaptive low-/high-frequency decomposition, specialised enhancement, adaptive fusion, and residual gated refinement. The SFNet-derived scaffold is used only for final optimisation and is not counted as a contribution.

- **Evidence.** We provide controlled three-domain evidence showing that the post-attention design improves a strong gated baseline and outperforms plausible alternatives, including pre-attention replacements and more complex routing/fusion variants. These results support the view that placement, rather than arbitrary frequency awareness or added complexity, is a first-order design choice for UIE restoration blocks.

2 Related Work

Underwater image enhancement. UIE has progressed from visual/physical priors to data-driven models trained on real paired data. UIEB and LSUI made supervised evaluation more systematic; Water-Net, Ucolor, and U-Shape Transformer respectively explored learned fusion, transmission/multi-colour cues, and global modeling for non-uniform attenuation [16, 17, 19]. Recent systems add real-time AUV correction, frequency-aware restoration, or physical constraints [3, 12, 25]. These works show that UIE needs scene context, structure preservation, and colour/visibility control. We are complementary: rather than proposing another full UIE pipeline or prior, we isolate a block-level question left implicit by Transformer-style UIE models–where frequency-aware repair should be placed inside an attention-FFN block.

Frequency-aware image restoration. Frequency reasoning is effective because degradations are not uniform across spectral bands. SFNet, FFTformer, and AdaIR use selective filtering, frequency-domain FFNs, or degradation-adaptive band modulation for restoration [10, 11, 14]; DEEP-SEA further links low/high processing with underwater structure preservation [3]. These works motivate frequency decomposition, but usually as a network-level branch, pipeline redesign, or fixed-location FFN. We ask a finer question: in a block with attention and two FFN slots, should frequency repair occur before or after global mixing?

Transformer block and FFN design for restoration. Restoration Transformers depend on more than attention: Restormer shows that gated depth-wise FFNs are active restoration operators, and FFTformer shows that FFNs can encode useful frequency biases [14, 23]. Prior inpainting work also demonstrates that placing FFNs on both sides of attention can expose two explicit intervention slots [2]. These works motivate stronger or frequency-aware FFNs, but rarely isolate placement. We use the two-FFN block only as a controlled testbed and ask whether frequency repair should happen before attention, where it will be re-mixed, or after attention, where global context is already available.

Datasets and evaluation. UIE evaluation is not a neutral reporting detail: paired references are scarce and benchmark "ground truth" is often a human-selected enhanced reference [4, 16, 19]. UIEB and LSUI make supervised UIE more systematic, but training mixtures, splits, references, and metric implementations still vary widely across papers [4, 16, 19]. We therefore use external numbers only as context and draw the placement conclusion from matched B-series comparisons. Metrics are interpreted jointly: PSNR/SSIM for fidelity [21], LPIPS for perceptual similarity [24], and UIQM/UCIQE as underwater no-reference indicators [18, 22], since the latter can diverge from visual quality [15].

3 Method

We instantiate the placement hypothesis in a two-FFN attention block, using its pre- and post-attention FFN slots as a controlled testbed rather than treating the block itself as the novelty. The key intervention is deliberately narrow: we leave the pre-attention FFN unchanged and replace only the post-attention FFN with SPARK. This replacement is designed around four sub-problems required by context-aware frequency repair: split the post-attention feature into low- and high-frequency components, repair the two bands according to their degradation roles, fuse them adaptively, and refine the fused representation into a stable restoration update. The remainder of this section first revisits the two FFN slots, then explains why the post-attention slot is preferred, details SPARK, and finally describes the network instantiation and training protocol.

3.1 Revisiting a two-FFN attention block

Given an input feature x, the block applies two FFNs around self-attention, each with its own layer normalization and residual connection:

$$x_1 = x + \mathrm{FFN}_{\mathrm{pre}}(\mathrm{LN}_1(x)), \tag{1}$$

$$x_2 = x_1 + \mathrm{Attn}(\mathrm{LN}_2(x_1)), \tag{2}$$

$$y = x_2 + \mathrm{FFN}_{\mathrm{post}}(\mathrm{LN}_3(x_2)). \tag{3}$$

This formulation is useful because it exposes two seemingly comparable intervention points inside the same restoration block: one before attention and one after attention. However, the two FFNs are not symmetric in function. The pre-attention FFN transforms features that have not yet been globally mixed, and its output is immediately consumed by self-attention. The post-attention FFN, in contrast, receives features after attention has aggregated information across the feature map and can therefore convert context-aware representations into the next restoration update. This asymmetry turns the block into a controlled testbed for asking where frequency-aware enhancement should be placed.

3.2 Why post-attention frequency enhancement?

The two FFN slots differ most in what information is available when they operate. A pre-attention frequency module acts on features before long-range context has been aggregated. Once its output enters self-attention, spatial activations are reweighted and re-mixed across the feature map, which is useful for context aggregation but can dilute band-specific enhancement before it becomes a stable restoration update. This is especially limiting for UIE, where low-frequency appearance errors such as colour bias, haze, and illumination drift should be interpreted with global scene evidence, while high-frequency structures such as edges and textures should be restored in relation to that global degradation state. The post-attention FFN is a better match to this requirement: it receives context-aware features and can convert the global view into separated low-frequency appearance repair and high-frequency structure repair. We therefore follow the principle *look globally, then repair by frequency*: keep the pre-attention FFN unchanged as a standard gated mixer, and place SPARK only at the post-attention slot.

3.3 SPARK: split, repair, fuse, and refine

Figure 2 is the method overview for the post-attention slot. The diagram should be read as one restoration update with four aligned actions, rather than as a list of disconnected layers. Given the context-aware feature x, SPARK *splits* it into low-frequency appearance and high-frequency structure states, *repairs* the two bands with asymmetric branches, *fuses* them with channel-aware adaptive fusion, and *refines* the fused cues through a residual GDFN body. This split–repair–fuse–refine path implements our principle: after attention has gathered global context, the FFN repairs by frequency.

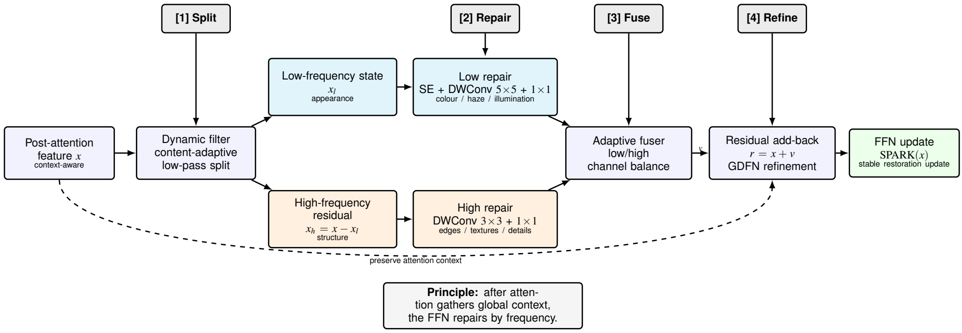

Figure 2: **SPARK as a post-attention split–repair–fuse–refine update.** Given the context-aware feature x after attention, SPARK first performs a content-adaptive low/high split, producing a low-frequency appearance state x_l and a high-frequency structure residual $x_h = x - x_l$. The low branch repairs colour, haze, and illumination cues, while the high branch repairs edges, textures, and fine details. An adaptive fuser balances the two repaired streams, and residual add-back plus GDFN refinement converts the fused cues into a stable post-attention FFN update.

Split: content-adaptive low/high decomposition. The first step is to make the two restoration roles explicit. Given the post-attention feature $x \in \mathbb{R}^{B\times C\times H\times W}$, we use the decomposition path of a dynamic filter to obtain a learned low-frequency response and define the remaining signal as the high-frequency residual:

$$x_l = \mathrm{DynFilt}_{\mathrm{dec}}(x), \qquad x_h = x - x_l. \tag{4}$$

Unlike a fixed blur kernel or a hand-designed Fourier mask, the dynamic filter predicts a content-adaptive low-pass kernel from globally pooled feature statistics, so the split can vary with the current underwater scene. We interpret x_l as the slow-varying appearance state and x_h as the residual structural evidence. The split itself is not the restoration result; it prepares a context-aware representation on which the following branches can perform different repairs.

Repair: specialised low- and high-frequency branches. After decomposition, the two bands are repaired with deliberately asymmetric lightweight branches. For the low-frequency state, we apply channel re-weighting followed by a larger depth-wise convolution and a point-wise projection:

$$u_l = \mathrm{Conv}_{1\times1}\big(\mathrm{DWConv}_{5\times5}(\mathrm{SE}(x_l))\big). \tag{5}$$

For the high-frequency residual, we use a tighter local branch:

$$u_h = \mathrm{Conv}_{1\times 1}\big(\mathrm{DWConv}_{3\times 3}(x_h)\big). \tag{6}$$

The asymmetry is intentional: low-frequency restoration benefits from broader channel and spatial adjustment, whereas high-frequency restoration should remain local and detail-preserving. Since both branches operate after attention, they do not need to infer the scene state from scratch; they translate context-aware low/high components into complementary restoration cues.

Fuse: adaptive recombination. The repaired streams should not be merged with a fixed sum, because the required balance between appearance correction and detail recovery varies across underwater scenes and feature channels. We therefore use the adaptive fuser to recombine u_l and u_h with channel-wise adaptive weights. Concretely,

$$[\alpha_l, \alpha_h] = \mathrm{Softmax}_c\big(\phi(\mathrm{GAP}([u_l, u_h]))\big), \qquad v = \mathrm{Conv}_{1\times 1}\big(\alpha_l \odot u_l + \alpha_h \odot u_h\big), \tag{7}$$

where ϕ denotes a lightweight projection used to produce the two channel-wise gates, and $\odot$ is channel-wise multiplication. This stage lets the post-attention FFN choose how much low-frequency appearance repair and high-frequency structure repair should contribute to the update, without imposing the same mixture on every image or channel.

Refine: residual connection and GDFN body. The fused feature v contains frequency-aware restoration cues, but it should not directly replace the post-attention representation. We treat it as a corrective update and add back the original context-aware input before applying a gated feed-forward body:

$$r = x + v, \qquad \mathrm{SPARK}(x) = \mathrm{GDFN}(r). \tag{8}$$

The residual connection preserves the information already organized by attention and reduces the risk of over-correcting when the low/high decomposition is uncertain. The GDFN body then provides the final nonlinear transformation, filtering the fused frequency cues and projecting them into the representation passed to the next block. Thus, the complete SPARK does not merely expose low- and high-frequency bands; it embeds those bands inside a complete residual feed-forward update.

3.4 Network instantiation and two-stage training protocol

The compact B-series UIE network used in this paper is our own implementation. It instantiates a four-level encoder–decoder restoration model with base width $d = 32$, block counts $[2,4,4,6]$, a global residual from the degraded RGB input to the decoder output, and a two-FFN attention block that exposes pre- and post-attention FFN slots. B2 uses gated FFNs at both slots; B3 keeps the pre-attention FFN gated and replaces only the post-attention FFN with SPARK; B4 and B7 test pre-attention alternatives; and B5 tests a different fusion design inside the post-attention module. Thus, the B-series architecture, two-FFN block instantiation, SPARK, and the B2/B3/B4/B5/B7 variants are implemented in our own codebase.

Our training workflow has two stages. Stage I uses this B-series implementation with a diagnostic training setup to run controlled FFN-slot ablations and select the architecture that answers the paper's placement question. Based on this stage, we select B3, where the pre-attention FFN remains gated and only the post-attention FFN is replaced by SPARK. Stage II then plugs the selected B3/SPARK network into an optimisation scaffold adapted from SFNet [10] for final training and performance reporting. This scaffold provides the optimiser, learning-rate schedule, random 256×256 crop, horizontal flip, multi-scale L_1 loss, and weighted FFT loss. It is used only as a training pipeline: the network module passed to the trainer is our implementation, no SFNet network architecture or pretrained weights are used in the reported models, and the SFNet training scaffold is not counted as a contribution.

4 Experiments

4.1 Unified protocol

The experiments are organised around the two-stage workflow described in §3. Stage I is the internal placement study: B2, B3, B4, B5, and B7 are trained and evaluated within the same diagnostic B-series setup, and the placement conclusion is drawn only from these within-stage controlled ablations. Stage II is the final optimisation stage: after B3 is selected, the same B3/SPARK architecture is trained with the SFNet-derived training scaffold for final performance reporting and external context. Cross-stage numbers are therefore not used to argue slot placement; they show the optimisation headroom of the selected architecture.

Data, metrics, and evaluation. All reported models use the same 4,800-pair training mixture drawn from EUVP, LSUI, and UIEB and the same evaluation script. Final reporting is conducted on three fixed test domains: EUVP with 200 pairs, LSUI with 200 pairs, and UIEB with 90 pairs. We report PSNR and SSIM for paired fidelity, LPIPS for perceptual similarity, and UIQM/UCIQE as underwater no-reference indicators. Stage-I comparisons use the matched diagnostic B-series setting, while Stage-II runs use the selected B3/SPARK network inside the optimisation scaffold adapted from SFNet, including Adam optimisation, warmup plus cosine decay, random crops, horizontal flips, multi-scale L_1 loss, and weighted FFT loss.

4.2 External comparison

Table 1 places our controlled cells next to representative UIE methods reported in the literature or re-tested by Guo et al. The table is deliberately separated from the internal ablation: its role is to show external context across EUVP, LSUI, and UIEB, while the placement claim is still drawn from matched B-series comparisons under our own protocol. In this broader view, the 100-epoch B3 gives the strongest UIEB PSNR among our internal cells, while the 650-epoch B3 improves EUVP/LSUI paired metrics within our runs and reaches the highlighted EUVP and LSUI PSNR/SSIM results; DEEP-SEA still reports a lower LSUI LPIPS under its author-reported protocol.

Table 1: **External comparison on EUVP / LSUI / UIEB.** PSNR↑ (dB) / SSIM↑ / LPIPS↓ / UIQM↑. AR = reported by original authors on their own splits; RT = re-tested by Guo et al. under a related benchmark. Our B3 rows use the selected B3/SPARK architecture; the 650-epoch row is Stage-II final training under the SFNet-derived training scaffold. External rows provide contextual evidence rather than strict protocol-matched head-to-head comparisons. "–" = not reported. Bold highlights the comparisons discussed in the text rather than automatically marking every column maximum; UIQM is not bolded because its absolute scale differs across implementations.

Method	Yr	Src	EUVP				LSUI				UIEB			
			PSNR↑	SSIM↑	LPIPS↓	UIQM↑	PSNR↑	SSIM↑	LPIPS↓	UIQM↑	PSNR↑	SSIM↑	LPIPS↓	UIQM↑
FUnIE-GAN [9]	'20	AR	21.92	0.888	–	2.78	24.23	0.858	0.182	–	19.45	0.85	–	1.03
Water-Net [16]	'20	RT	24.12	0.839	0.222	2.99	25.25	0.877	0.164	3.08	22.82	0.907	0.125	3.26
Ucolor [17]	'21	RT	23.72	0.828	0.205	3.05	21.30	0.821	0.225	3.06	18.37	0.814	0.221	3.04
PUIE-Net [6]	'22	RT	20.48	0.784	0.270	3.20	22.07	0.864	0.191	3.53	21.04	0.877	0.136	3.54
Semi-UIR [8]	'23	RT	24.59	0.821	0.172	2.94	25.40	0.843	0.160	3.05	23.64	0.888	0.120	3.17
PUGAN [5]	'23	AR	24.05	–	–	2.94	–	–	–	–	21.67	–	–	3.28
U-Shape Trans. [19]	'23	RT	24.99	0.829	0.238	2.92	25.15	0.838	0.221	3.05	20.75	0.810	0.228	3.11
DeepWaveNet [20]	'23	RT	23.41	0.836	0.204	3.01	23.81	0.870	0.177	3.06	21.55	0.904	0.143	3.13
DEEP-SEA [3]	'25	AR	29.27	0.905	0.131	–	28.90	0.897	0.090	–	–	–	–	–
B2 (gated/gated baseline)	–	ours	29.36	0.906	0.121	1.59	28.17	0.918	0.108	1.51	22.29	0.902	0.137	1.99
B3 (SPARK, 100 ep)	–	ours	29.33	0.906	0.123	1.60	28.11	0.919	0.110	1.52	**23.11**	0.913	0.123	1.97
B3 (SPARK, 650 ep)	–	ours	**30.19**	**0.915**	**0.108**	1.78	**29.02**	**0.925**	**0.097**	1.53	22.55	0.912	0.127	2.03

Protocol-comparability caveat. Table 1 should be read as ordering evidence rather than as a precise dB-level head-to-head. RT numbers are re-tested by Guo et al. [7] under a 90/200/200 UIEB/EUVP/LSUI benchmark; this is close to our 200/200/90 split, but the image-level lists may differ. AR numbers come from original authors on their own splits, such as the 1K EUVP subset for FUnIE-GAN, Test-L400 for U-Shape, and Test-EUVP for PUGAN. We omit UCIQE from Table 1 because our implementation uses the unnormalised Yang–Sowmya formula [22], whereas prior UIE work often uses a $[0,1]$-normalised variant; UIQM may also differ in scale across implementations. UCIQE is therefore kept only in internal comparisons where all cells share our implementation.

4.3 Main controlled result: B2 vs B3

The first controlled question is whether the proposed post-attention FFN improves a strong baseline within Stage I. Table 2 therefore separates the matched Stage-I placement ablation from the Stage-II longer B3 training run. Under the primary B2→B3 comparison, the three-domain mean PSNR increases from 26.60 to 26.85, SSIM from 0.9089 to 0.9126, LPIPS decreases from 0.1222 to 0.1185, and UCIQE increases from 27.59 to 27.73. The absolute mean gain is modest, and we interpret it as a diagnostic slot-level effect rather than as a large SOTA-style jump; its significance comes from the controlled setting, the consistent multi-metric direction, and the fact that higher-capacity or more complex alternatives do not reproduce the same gain.

Table 2: **Internal ablation under our B-series implementation.** The top block is the Stage-I internal placement ablation, where B2/B3/B4/B5/B7 are compared within the same diagnostic setup to test where frequency-aware repair should be placed. The bottom block reports Stage-II final training behaviour of the selected B3/SPARK architecture under the SFNet-derived training scaffold. The placement conclusion is drawn from the Stage-I controlled variants; the Δ row reports B3-650ep minus B3-100ep in the Stage-II block.

Cell	Params	PSNR↑	SSIM↑	LPIPS↓	UIQM↑	UCIQE↑
Stage-I internal placement ablation: where to put SPARK						
B2 (gated / gated)	24.09M	26.60	0.9089	0.1222	1.6970	27.59
B4 (mres pre / SPARK)	29.18M	26.63	0.9102	0.1199	1.6670	27.66
B5 (gated / SPARK-CS)	28.59M	26.72	0.9114	0.1186	1.7003	27.64
B7 (freqMoE pre / SPARK)	31.71M	26.65	0.9109	0.1198	**1.7190**	27.64
B3 (gated / SPARK)	28.59M	**26.85**	**0.9126**	**0.1185**	1.6980	**27.73**
Stage-II final training under the SFNet-derived scaffold						
B3 (100 ep)	28.59M	26.85	0.9126	0.1185	1.6980	**27.73**
B3 (650 ep)	28.59M	**27.25**	**0.9175**	**0.1107**	**1.7810**	27.62
Δ (650 vs 100)	–	+0.40	+0.005	-0.008	+0.083	-0.11

The Stage-II long-training rows should be read separately from the placement claim. They use the selected B3 architecture inside the SFNet-derived training scaffold and show that the post-attention design benefits from more optimisation, improving mean PSNR by +0.40 dB and LPIPS by -0.008 over B3-100ep, while slightly reducing UCIQE. We therefore use the Stage-I rows for the apples-to-apples slot ablation and use B3-650ep only as headroom evidence for the selected architecture.

4.4 Domain-wise behaviour and metric trade-offs

The improvement from B2 to B3 is not uniform across domains. In Table 1, the 100-epoch B3 stays close to B2 on EUVP and LSUI, but improves UIEB PSNR from 22.29 to 23.11. The same UIEB pattern appears in the full-reference metrics: SSIM increases from 0.902 to 0.913 and LPIPS decreases from 0.137 to 0.123. This concentration of gains is consistent with our motivation: UIEB exposes difficult real-scene cases where large-scale colour and haze correction must be balanced with local structure preservation. The Stage-II 650-epoch B3 row improves EUVP and LSUI more strongly, suggesting that the same selected architecture also benefits from the SFNet-derived optimisation scaffold, but the placement claim remains the Stage-I 100-epoch B2/B3 comparison.

Paired significance. We additionally run no-retrain paired tests on matched per-image outputs for B2, B3, B5, and B7. B3 is significantly better than B2 on UIEB in four of five metrics: PSNR +0.82 dB (95% CI [0.28, 1.42], $p = 0.009$), SSIM +0.011 ($p = 0.042$), LPIPS -0.014 ($p = 0.008$), and UCIQE +0.42 ($p = 0.032$); UIQM is not significant. On EUVP and LSUI, the B3–B2 confidence intervals include zero, so we treat those changes as within noise. This tightens the claim: the post-attention design is most useful on the real-scene UIEB domain where the appearance–structure trade-off is most exposed.

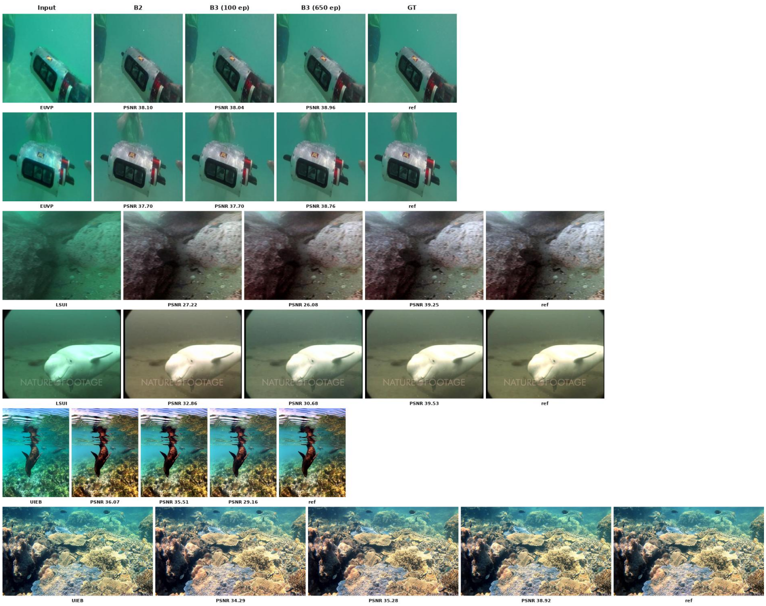

Figure 3: **Qualitative comparison on EUVP, LSUI, and UIEB.** Columns are input, B2, B3 (100 ep), B3 (650 ep), and GT. The examples visualise the Table 1 trend: long training improves EUVP/LSUI, while the 100-epoch B3 remains important on UIEB.

The metric view also clarifies how to read the ablations. B3 gives the best 100-epoch three-domain mean on PSNR, SSIM, LPIPS, and UCIQE, while B7 is slightly higher on UIQM. We therefore treat B7's UIQM lead as a trade-off in a no-reference metric, not as evidence against the placement conclusion: B3 is preferred because it gives the strongest balance across paired fidelity, perceptual similarity, and underwater-specific indicators under the matched protocol.

4.5 Placement ablation: why pre-attention variants fail

The top block of Table 2 should be read as a set of counterfactuals, not as a separate leaderboard. B4 adds more pre-attention capacity (+5.09M parameters over B2), B7 makes the pre-attention slot frequency-aware with a top-1 low/high MoE and is larger than B3 (31.71M vs. 28.59M), and B5 is parameter-matched to B3 while making the post-attention fuser more fine-grained. They reach 26.63, 26.65, and 26.72 mean PSNR, respectively, all below B3 at 26.85. Thus, neither model size, pre-attention frequency routing, nor fuser complexity explains the B3 gain. This makes the placement hypothesis falsifiable: if the improvement came merely from added capacity, frequency awareness, or finer routing, then B4, B7, or B5 should match B3 under the same protocol; none of them does.

Why pre-attention modification fails. The reason is structural. A pre-attention frequency module operates before global context is collected, and its output is immediately passed into self-attention, where spatial activations are reweighted and re-mixed across the feature map. Any band-specific repair introduced there can therefore be diluted before it becomes a stable restoration update. This is a poor match for UIE: colour cast, haze, and illumination drift should be interpreted with scene-level evidence, while edges and textures should be restored relative to that global degradation state. The post-attention FFN has the opposite property: it receives context-aware features and directly converts them into low-frequency appearance repair and high-frequency structure repair. The failures of B4, B5, and B7 are therefore not negative results; they are the evidence that placement matters.

4.6 Internal ablation: residual + GDFN body

The refine-stage ablation in Table 3 asks a different question from the placement study: after low/high cues have been generated, what is required inside the post-attention slot? The simplified `spark_post` cell keeps the dynamic-filter decomposition and adaptive low/high fusion, but removes the residual add-back and final GDFN body. This is not sufficient: its mean PSNR is 26.57, slightly below the B2 baseline at 26.60. Restoring the full residual + GDFN refinement produces B3, reaching 26.85 mean PSNR and 23.11 on UIEB. Thus, the full module is effective not because it merely exposes frequency bands, but because it embeds them in a complete residual feed-forward update.

Table 3: **Stage-I diagnostic ablation of the refine stage.** `spark_post` keeps the content-adaptive low/high split and adaptive fusion, but removes the residual add-back and GDFN body. Frequency cues alone are not sufficient; full B3 embeds them in a stable post-attention FFN update and gives the best mean.

Cell	Params	EUVP	LSUI	UIEB	Mean
B2 (gated/gated baseline)	24.09M	**29.36**	**28.17**	22.29	26.60
`spark_post`	27.45M	29.12	28.14	22.46	26.57
B3 (SPARK)	28.59M	29.33	28.11	**23.11**	**26.85**

4.7 Limitations and failure cases

The controlled results also define the boundary of our claim. SPARK is not intended to solve every UIE setting, and the current evidence should be read as a slot-level architectural finding rather than a universal benchmark claim. LSUI remains challenging in the matched 100-epoch comparison: using the reported DEEP-SEA EUVP/LSUI values (29.27/28.90), B3-100ep is 0.06 dB higher on EUVP but 0.79 dB lower on LSUI. The Stage-II B3-650ep run narrows this gap and slightly exceeds DEEP-SEA on LSUI PSNR, but it is not a Stage-I placement ablation and should be read as optimisation headroom rather than the basis of the placement claim. There are also practical limitations. The larger variants require reduced batch sizes and activation checkpointing within 32 GB memory. A no-retrain complexity benchmark at 256×256 shows that B3 increases B2 from 24.09M to 28.59M parameters, 63.84 to 71.22 GFLOPs, and 21.17 to 27.74 ms/image; B7 is larger and slower (31.71M, 76.29 GFLOPs, 33.24 ms/image) but remains below B3 in mean PSNR. Thus, the placement gain is not simply a larger-model effect. Because the current comparison is single-seed, we treat the ordering as diagnostic evidence for a design hypothesis rather than as a statistical performance claim; multi-seed evaluation remains necessary to quantify variance.

5 Conclusion

This paper studies where frequency repair should happen inside a Transformer-style UIE block. Instead of treating frequency decomposition as location-free, we use the two FFN slots around attention to test the principle *look globally, then repair by frequency*. SPARK redesigns only the post-attention FFN with a compact split–repair–fuse–refine update. Stage-I controlled ablations show that this slot-specific intervention improves a strong gated baseline and outperforms alternatives that add capacity, pre-attention frequency routing, or more complex fusion; Stage-II training with the SFNet-derived scaffold shows optimisation headroom for the selected architecture. The conclusion is diagnostic: frequency awareness is insufficient unless placed where the block can use global context.

This conclusion is strongest as a controlled B-series comparison, not a complete benchmark claim across all UIE models. Future work should test the placement principle under multi-seed training, broader evaluation splits, and additional real-world domains, and combine post-attention frequency repair with stronger domain generalization or physically grounded degradation modeling.

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